



PAAVAI COLLEGE OF ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

An AI-Driven Intelligent Traffic Signal and Safety Management System Using Deep Reinforcement Learning

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AI-POWERED SMART TRAFFIC MANAGEMENT SYSTEM

Real-time, adaptive control with YOLOv8.



EMERGENCY VEHICLE DETECTED

– Priority Override Active

Lane 1 - Density: 26.9 | Veh. 20

Vehicle Count: 20
Density: 26.9



YOLOv8

Lane 3 - Density: 25.9 | Veh. 24

Vehicle Count: 24
Density: 25.9



YOLOv8

Lane 2 - Density: 25.9 | Veh. 24

Vehicle Count: 24
Density: 25.9



YOLOv8

Lane 4 - Density: 26.9 | Veh. 20

Vehicle Count: 20
Density: 26.9



YOLOv8



1. Real-time Monitoring



2. AI Vehicle Detection



3. Classification



4. Vehicle Counting



5. Emergency Detection



6. Analysis & Reporting



7. Web-based Dashboard

Abstract



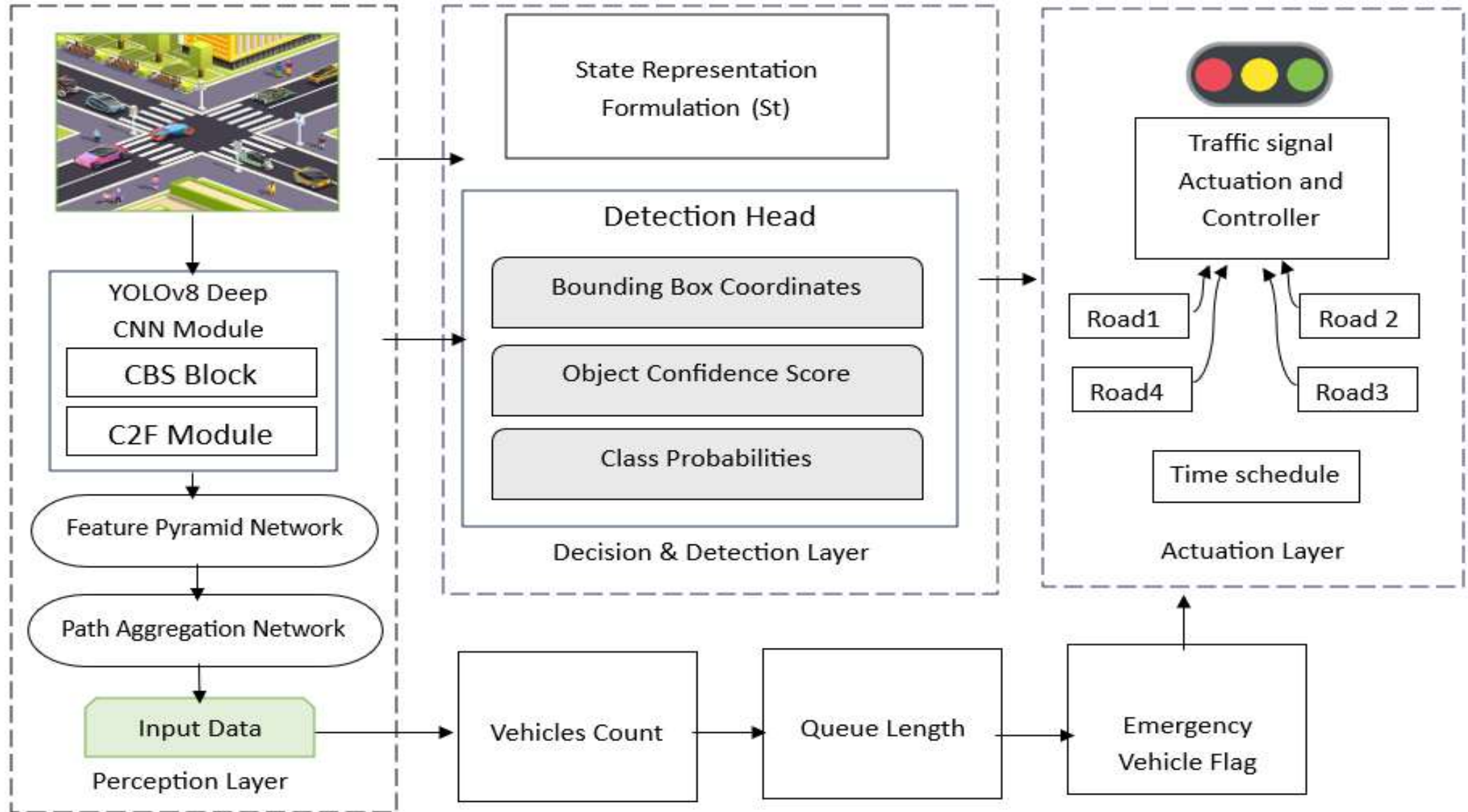
- The Development of urban Vehicle Traffic System necessitates Artificial Intelligence based solutions beyond the convolution pre-timed controlling systems.
- These intelligence systems often lead to unnecessary waiting time in crowd, fuel consumption and accidents.
- This paper presents a intelligence based Traffic Controlling System that leverages computer vision for analysing vehicles and Deep Reinforcement Learning (DRL) for adaptive light optimization.
- The framework utilizes YOLOv8, A State-of-Art convolutional Neural Network (CNN) model, for analysing real-time CCTV footage from four interesting roads.



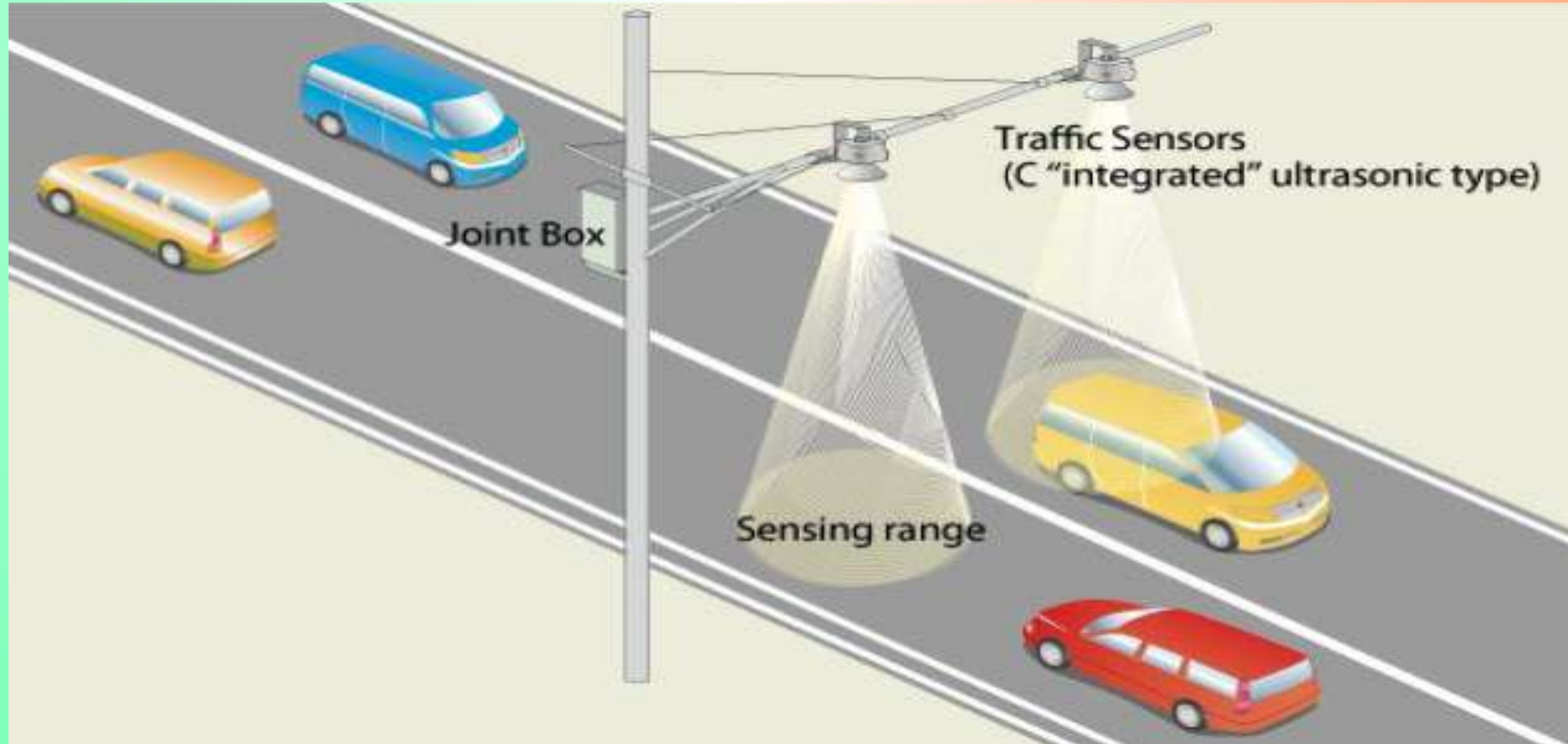
PROBLEM STATEMENT

- The rapid growth of urban smart cities has leads to increase in traffic on roads, making traffic pollution, increased waiting time on traffic roads, which challenges in congestion in metropolitan areas worldwide.
- Conventional traffic signal control systems, which works on fixed-time are fundamentally lacking for managing the dynamic and non-uniform nature of modern traffic flows.
- These integrated systems lack the perceptual capability to react to real-time changes in traffic crowds, unexpected traffic Jams, the presence of emergency vehicles in crowd locations, often results in increased vehicle waiting times, fuel wastage and elevated environmental pollutions

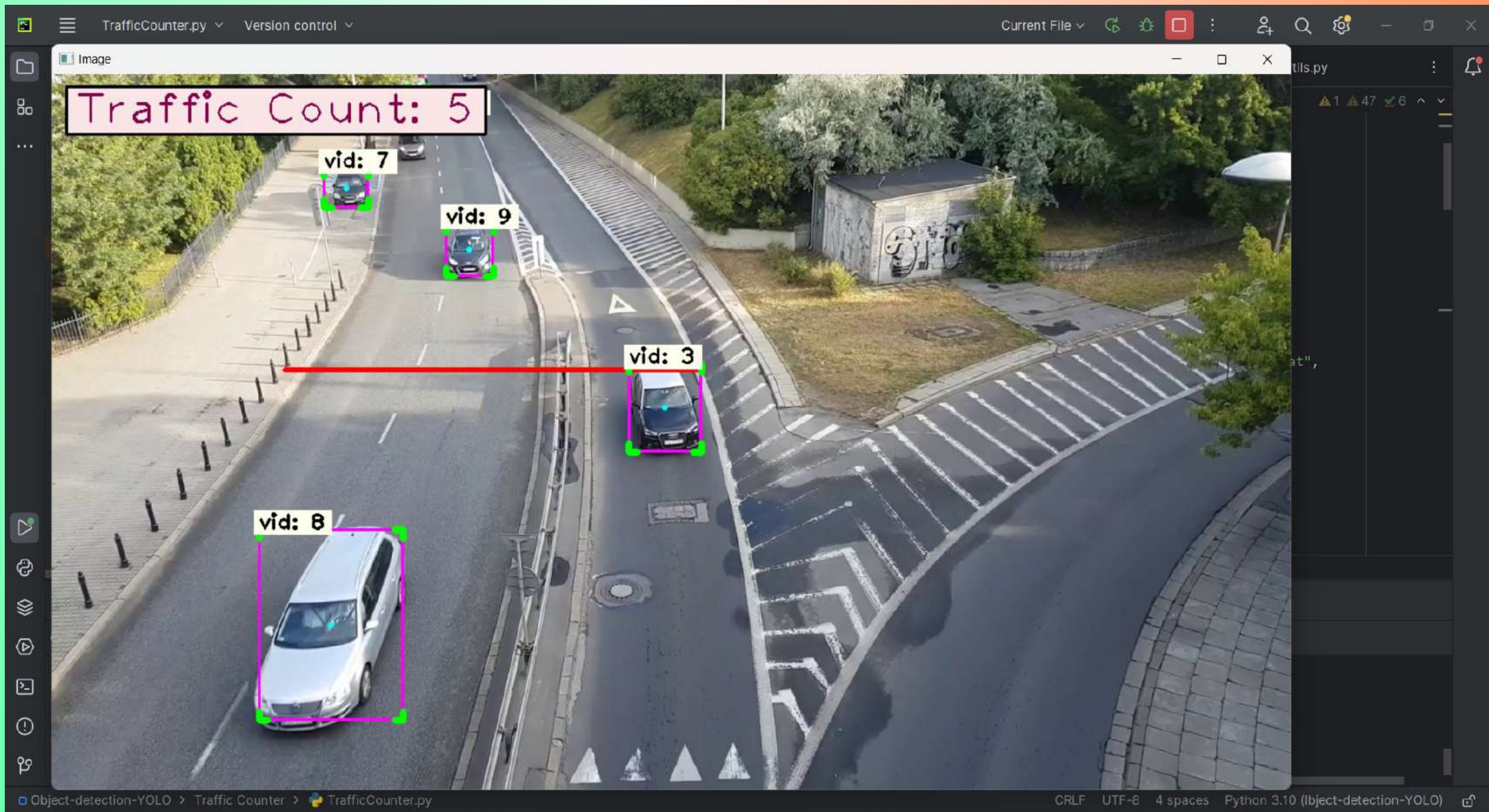
System Architecture for AI driven Intelligent Traffic Control System



Process



Real-Time Traffic Monitoring



AI-Based Vehicle Detection

AI-Based SMART TRAFFIC SYSTEM

On-Screen Results:

Vehicles Detected: 15

Emergency Vehicle: Yes

Signal Decision: **GREEN**
(Emergency Priority)

Ambulance

Fire Truck

Car

Car

Car

Truck

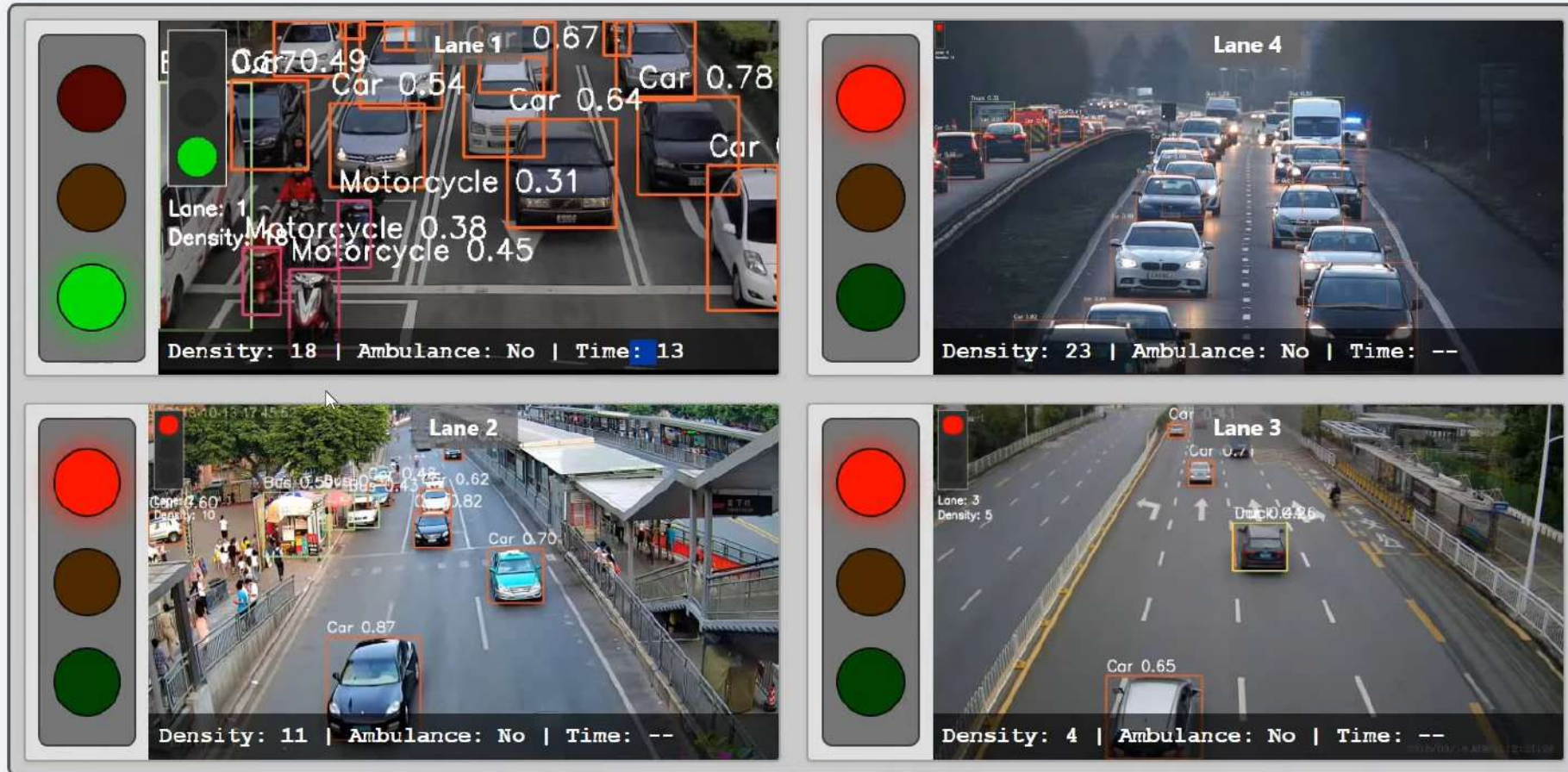
Car

Car



Vehicle Classification

Live Traffic Dashboard



Automated Vehicle Counting

```
C:\Windows\system32\cmd.e: X + v

127.0.0.1 - - [28/Oct/2025 17:08:06] "GET /status_api HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2025 17:08:07] "GET /status_api HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2025 17:08:08] "GET /status_api HTTP/1.1" 200 -
Lane 3 turning red. Added {'Car': 0, 'Bus': 0, 'Truck': 0, 'Motorcycle': 0, 'Ambulance': 0} to cumulative total.
Switching light: Next Lane 4 (Green)
Lane 4 is Green for 5s (Density: 0)
127.0.0.1 - - [28/Oct/2025 17:08:09] "GET /status_api HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2025 17:08:10] "GET /status_api HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2025 17:08:11] "GET /status_api HTTP/1.1" 200 -
Lane 1 is now Orange for 3s
127.0.0.1 - - [28/Oct/2025 17:08:12] "GET /status_api HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2025 17:08:13] "GET /status_api HTTP/1.1" 200 -
Lane 4 is now Orange for 3s
127.0.0.1 - - [28/Oct/2025 17:08:14] "GET /status_api HTTP/1.1" 200 -
Lane 1 turning red. Added {'Car': 11, 'Bus': 0, 'Truck': 2, 'Motorcycle': 1, 'Ambulance': 0} to cumulative total.
Switching light: Next Lane 2 (Green)
Lane 2 is Green for 17s (Density: 12)
127.0.0.1 - - [28/Oct/2025 17:08:15] "GET /status_api HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2025 17:08:16] "GET /status_api HTTP/1.1" 200 -
Lane 4 turning red. Added {'Car': 0, 'Bus': 0, 'Truck': 0, 'Motorcycle': 0, 'Ambulance': 0} to cumulative total.

--- Full cycle complete. Restarting from Lane 1. ---
Switching light: Next Lane 1 (Green)
Lane 1 is Green for 5s (Density: 0)
127.0.0.1 - - [28/Oct/2025 17:08:17] "GET /status_api HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2025 17:08:18] "GET /status_api HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2025 17:08:19] "GET /status_api HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2025 17:08:20] "GET /status_api HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2025 17:08:21] "GET /status_api HTTP/1.1" 200 -
Lane 1 is now Orange for 3s
127.0.0.1 - - [28/Oct/2025 17:08:22] "GET /status_api HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2025 17:08:23] "GET /status_api HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2025 17:08:24] "GET /status_api HTTP/1.1" 200 -
Lane 1 turning red. Added {'Car': 0, 'Bus': 0, 'Truck': 0, 'Motorcycle': 0, 'Ambulance': 0} to cumulative total.
Switching light: Next Lane 2 (Green)
Lane 2 is Green for 5s (Density: 0)
127.0.0.1 - - [28/Oct/2025 17:08:25] "GET /status_api HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2025 17:08:26] "GET /status_api HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2025 17:08:27] "GET /status_api HTTP/1.1" 200 -
127.0.0.1 - - [28/Oct/2025 17:08:28] "GET /status_api HTTP/1.1" 200 -
```

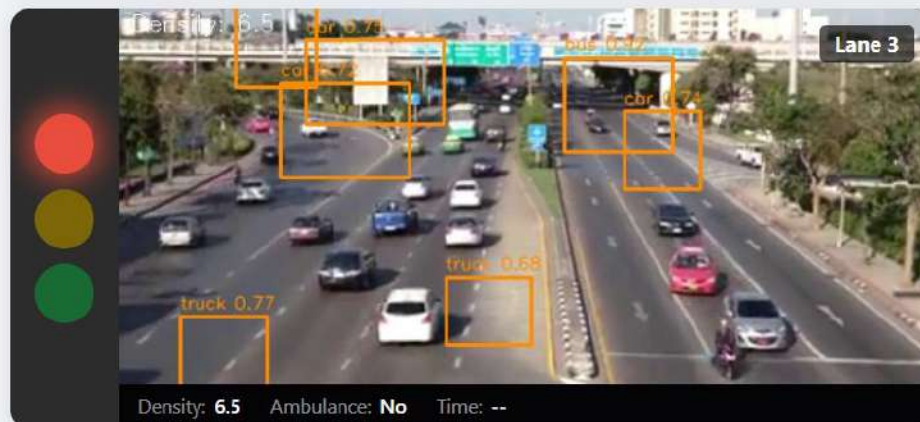
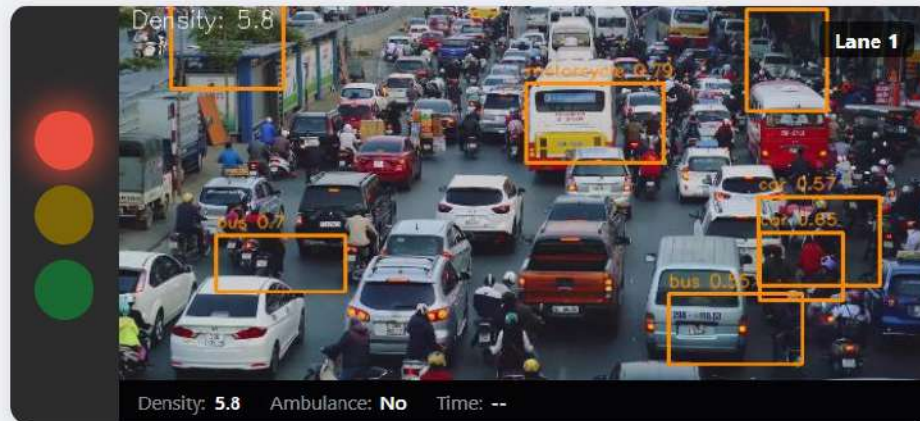

Emergency Vehicle Detection



ambulance: 95%

Traffic Analytics and Reporting

✓ Analysis Complete



Web-Based Monitoring Dashboard

Intelligent Traffic Management System

Using Deep Learning and YOLOv8 to optimize traffic flow, prioritize emergency vehicles, and provide detailed analytics.

Admin Login

⚠ Invalid username or password.

Username

traffic-admin

Password

➡ Login



Traffic AI Dashboard

Adaptive signal control powered by YOLOv8 deep learning.

Upload 4 lane videos to begin real-time traffic analysis.



4

LANES



YOLOv8

AI MODEL



Live

EMERGENCY



Real-time

ANALYSIS



Start Project



Vehicle Detection



Emergency Priority



Density Analysis



Parallel Processing

Live Traffic Dashboard

TensorFlow + YOLOv8

Analyzing lanes...

EMERGENCY VEHICLE DETECTED — Priority Override Active





Vehicles: 21 Density: 28.2 Ambulance: YES Color Detection 1.00 Signal: RED

Time: --



Vehicles: 21 Density: 28.2 Ambulance: YES Color Detection 1.00 Signal: RED

Time: --

Signal Decision Log

Lane	Vehicles	Density	Emergency	Detection Method	Signal	Green Time	Reason
Lane 1	21	28.2	YES	Color Detection	GREEN	60	Multi-Emergency — Highest density (28.2)
Lane 4	21	28.2	YES	Color Detection	RED	--	Emergency — waiting (lower density)
Lane 2	21	28.2	YES	Color Detection	RED	--	Emergency — waiting (lower density)
Lane 3	21	28.2	YES	Color Detection	RED	--	Emergency — waiting (lower density)

New Analysis

Future Enhancement

- Multi-Junction Control
- Edge Computing
- Autonomous Vehicle Support
- IoT Sensor Integration
- Weather Awareness
- Pedestrian Detection
- Accident Detection

Conclusion

We envision a fully coordinated and resilient urban network.

Our future work will pursue three parallel paths to realize Multi-Agent Coordination, All-Weather Perception, Digital Twin Deployment, These steps are designed to transition the technology from a proof-of-concept at one intersection to a scalable, robust foundation for the smart city infrastructure of tomorrow.

THANK YOU